

GUSTAVO RODRIGUEZ

Mechanical Engineering Student | Expected Graduation: December 2028
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RESEARCH & ENGINEERING LEADERSHIP

IEEE EXO, University of South Florida

Tampa, FL

Project Lead / EXO Chair and President

August 2025 – Present

- Lead a multidisciplinary lower-limb exoskeleton program spanning mechanical design, electrical, controls, research, fabrication, testing, and safety; manage design reviews, technical deliverables, procurement, subsystem dependencies, and approximately \$5,000 in development budget.
- Define requirements for a unilateral hip/knee assistive prototype, including joint alignment, human interfaces, actuator sizing, sensing, power, controls, and test planning; guide integration of AK80-9 actuators, IMUs, encoders, Jetson Orin Nano, and Teensy 4.1 controllers.
- Translate biomechanics and motion-analysis goals into engineering decisions on assistance timing, torque targets, prototype architecture, safety limits, and validation strategy while coordinating cross-team milestones and technical documentation.

CARRT Lab, University of South Florida

Tampa, FL

REU Research Assistant

August 2026 – Present

- Analyze IMU and Vicon motion-capture data and support experimental setup, participant data collection, sensor validation, post-processing, data-quality checks, and technical documentation for human-motion and biomechanics research.

ENGINEERING EXPERIENCE

Lonza

Tampa, FL

Reliability Engineering Intern

August 2026 – December 2026

- Investigate recurring Xcelodose 600S and microbalance failures through physical inspection, mechanical alignment, spring/load-cell geometry, corner-load behavior, SOP-to-practice comparison, and equipment-condition review; use 6M, Gemba, and FMEA/failure-mode analysis to isolate likely failure pathways.
- Compare symptoms across multiple failed units to distinguish mechanical, sensing, communication, and software-interface issues; translate findings into corrective actions, operator guidance, controlled documentation updates, and recurrence-prevention training.
- Support Engineering, Reliability, Manufacturing, and Project Engineering with equipment-risk reviews, validation/readiness activities, technical-record review, process bottleneck investigations, and project-equipment decisions in a regulated manufacturing environment.

Facilities, Engineering & Maintenance Intern

May 2026 – August 2026

- Supported preventive maintenance, equipment inspections, HVAC/AHU work, troubleshooting, work orders, SOP updates, technician follow-up, and contractor/facility coordination across manufacturing facilities.
- Led a Lean/5S A3 asset-identification improvement using standardized high-visibility labels and before/after time trials, reducing average equipment search time from roughly 9 minutes to 1.5 minutes (83%).
- Built digital maintenance and manufacturing workflows using Power Apps, Power Automate, SharePoint/Lists, Forms, and Excel, including QR-linked cleaning-status tracking and structured operator/process records for better traceability and shift continuity.
- Developed technician-facing maintenance training around Receive → Prepare → Execute → Test/Verify → Document → Close/Handoff; incorporated GDP/ALCOA, escalation, work-order, and return-to-service expectations.
- Supported five-year lifecycle/capital planning by cleaning and cross-referencing condition, risk, cost, ownership, and replacement data for 300+ high-level equipment and infrastructure assets.

TECHNICAL SKILLS & CREDENTIALS

Engineering: SolidWorks 2025, CAD/assembly modeling, drawing/specification review, FDM 3D printing, equipment validation, RCA/FMEA

Software / Hardware: Python, MATLAB, Excel, Power Apps/Automate, Arduino/ESP32, Jetson/Teensy, sensors and actuators

Manufacturing / Quality: Lean Six Sigma, 5S/A3, GMP/GDP, SOPs, visual management, lifecycle/CapEx support

Credentials / Languages: Lean Six Sigma Yellow Belt; CSWA - CAD Design; CSWA - Sustainability; ReCF Pre-Engineering & Robotics; CITI Biomedical/Research Integrity; Altium Designer introductory workshop series (in progress); English, Spanish

SELECTED TECHNICAL PROJECTS

S.M.U.R.F. - Shared Maintenance Urgent Request Form | Power Apps, Power Automate, SharePoint/Lists

- Built a two-app Power Apps/Automate maintenance workflow with five concern-specific decision branches, automating urgency classification, HTML alerts/reminders, visual prioritization, assignment tracking, and searchable records.

Mime Robotic Hand | Python, MediaPipe, ESP32, PCA9685, Servo Control

- Integrated computer vision with an open-source 5-DOF tendon-driven hand; developed finger calibration, mapping, smoothing/hysteresis, reverse-servo logic, and a Rock-Paper-Scissors interaction mode.

Autonomous Color-Following Robot | SolidWorks, Arduino Uno, ESP32, Pixy2, 3D Printing

- Designed and fabricated a custom 3D-printed chassis and integrated vision sensing, embedded control, low-voltage wiring, autonomous following, LCD feedback, and RC mode switching into a functional mobile robot.

EDUCATION

University of South Florida

Tampa, FL

Bachelor of Science in Mechanical Engineering, GPA: 3.33

Expected December 2028

- Coursework: Dynamics, Mechanics of Solids, Thermodynamics, Materials Engineering, Electrical Systems, CAD & Engineering.
- Recognition: Arthrex Scholar, USF Scholars Gold Recognition, Florida Academic Scholars.